CH 2 — Limits

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Limits

2.1 Definition of Limits

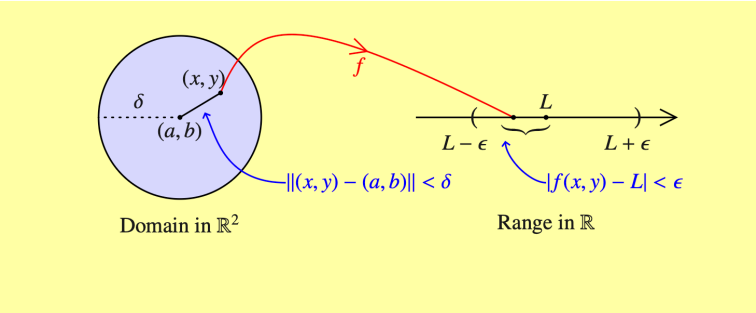
> 📘 **Definition — Neighborhood**
>
> An r-**neighborhood** of a point $(a, b) \in \mathbb{R}^2$ is a set
>
> $$N_r(a, b) = \{(x, y) \in \mathbb{R}^2 : \|(x, y) - (a, b)\| < r\}$$

Note:

$\|(x, y) - (a, b)\|$ is the **Euclidean distance** or $\sqrt{(x-a)^2 + (y-b)^2}$

Visualize that neighborhood as an area of (x, y) that is the interval for a 3D plane

> 📘 **Definition — Limits**
>
> Assume that $f(x, y)$ is defined in some neighborhood of (a, b), except possible at (a, b). Then if for every $\varepsilon > 0, \exists \delta > 0$ s.t.
>
> $$0 < \|(x, y) - (a, b)\| < \delta \longrightarrow |f(x, y) - L| < \varepsilon$$
>
> We say that $\lim_{(x,y)\to(a,b)} f(x, y) = L$



2.2 Limit Theorems

Limit Theorems

If $\lim_{(x,y)\to(a,b)} f(x, y)$ and $\lim_{(x,y)\to(a,b)} g(x)$ both exsits, then:

1. $\lim_{(x,y)\to(a,b)}[f(x, y) \pm g(x, y)] = \lim_{(x,y)\to(a,b)} f(x, y) \pm \lim_{(x,y)\to(a,b)} g(x, y)$
2. $\lim_{(x,y)\to(a,b)}[f(x, y)g(x, y)] = \left[\lim_{(x,y)\to(a,b)} f(x, y)\right]\left[\lim_{(x,y)\to(a,b)} g(x, y)\right]$
3. $\lim_{(x,y)\to(a,b)} \frac{f(x,y)}{g(x,y)} = \frac{\lim_{(x,y)\to(a,b)} f(x,y)}{\lim_{(x,y)\to(a,b)} g(x,y)}$, for any $\lim_{(x,y)\to(a,b)} g(x, y) \neq 0$
4. If $\lim_{(x,y)\to(a,b)} f(x, y)$ exists, then it is also unique

Proof

Uniqueness of a limit

Assume that $\lim_{(x,y)\to(a,b)} f(x,y) = L_1$ and $\lim_{(x,y)\to(a,b)} f(x,y) = L_2$
$L_1 - L_2 = \lim_{(x,y)\to(a,b)} f(x,y) - \lim_{(x,y)\to(a,b)} f(x,y) = \lim_{(x,y)\to(a,b)}[f(x,y) - f(x,y)]$
$\implies \lim_{(x,y)\to(a,b)} 0 = 0 \implies L_1 - L_2 = 0 \implies L_1 = L_2$

□

2.3 Proving a Limit denoted

In $\mathbb{R}^2$ we can approach the point (a,b) along an infinite number of paths.

For the limit to exsits, f must approach L along **ALL** paths

Examples:

1. Let $f(x,y) = \frac{y^2-x^2}{y^2+x^2}$, prove $\lim_{(x,y)\to(0,0)} f(x,y) = \text{DNE}$

   Try approaching $(0,0)$ along straight lines of the form $y = mx$

   Along the line $y = mx$,

   $\lim_{(x,y)\to(0,0)} f(x,y) = \lim_{x\to 0} f(x,mx) = \lim_{x\to 0} \frac{m^2x^2-x^2}{m^2x^2+x^2} = \lim_{x\to 0} \frac{m^2-1}{m^2+1} = \frac{m^2-1}{m^2+1}$

   f approaches different values depending on m, so the limit does not exists as it is not unique.

2. Let $f(x,y) = \frac{|x|}{|x|+y^2}; (x,y) \neq (0,0)$, show that $\lim_{(x,y)\to(0,0)} f(x,mx) = 1 \ \forall m \in \mathbb{R}$, but $\lim_{(x,y)\to(0,0)} f(x,y) = \text{DNE}$

   Try approaching $(0,0)$ along straight lines of the form $y = mx$

   Along the line $y = mx$,

   $\lim_{(x,y)\to(0,0)} f(x,y) = \lim_{x\to 0} f(x,mx) = \lim_{x\to 0} \frac{|x|}{|x|+m^2x^2}$

   $\implies \lim_{x\to 0^-} \frac{-x}{-x+m^2x^2} = \lim_{x\to 0^-} \frac{-1}{-1+m^2x} = 1$

   $\implies \lim_{x\to 0^+} \frac{x}{x+m^2x^2} = \lim_{x\to 0^+} \frac{1}{1+m^2x} = 1$

   $\lim_{(x,y)\to(0,0)} f(x,y) = 1$

   Try approaching $(0,0)$ along $x = 0$

   $\lim_{(x,y)\to(0,0)} f(x,y) = \lim_{x\to 0} f(0,y) = 0 \neq 1$

Note: the above approach cannot be used to prove the existence of limit.

3. Show that $\lim_{(x,y)\to(0,0)} \frac{x^2y}{x^4+y^2}$ does not exist. (**Exponent matching**)

   Try approaching $(0,0)$ along the straight lines of the form $y = mx$

   $\lim_{(x,y)\to(0,0)} \frac{x^2(mx)}{x^4+(mx)^2} = \lim_{(x,y)\to(0,0)} \frac{mx}{x^2+m^2} = 0, \ \forall (x,y) \neq (0,0)$

   Try approaching $(0,0)$ along $y = x^2$

   $\lim_{(x,y)\to(0,0)} \frac{x^2(x^2)}{x^4+x^4} = \lim_{(x,y)\to(0,0)} \frac{1}{2} = \frac{1}{2}, \ \forall (x,y) \in \mathbb{R}^2$

   Notice that $0 \neq \frac{1}{2}$. Thus the limit does not exists.

📐 **Theorem — Squeeze Theorem**

If there exists a function $B(x,y)$ s.t.

$$|f(x,y)-L| \leq B(x,y),\ \ \forall (x,y) \neq (a,b)$$

in some neighborhood of (a,b) and

$$\lim_{(x,y)\to(a,b)} B(x,y) = 0$$

then

$$\lim_{(x,y)\to(a,b)} f(x,y) = L$$

Proof

Let $\varepsilon > 0$.

Since $\lim_{(x,y)\to(a,b)} B(x,y) = 0$, $\exists \delta > 0$ s.t. $0 \leq \|(x,y)-(a,b)\| \leq \delta \longrightarrow |B(x,y)| < \varepsilon$

So, whenever $0 \leq \|(x,y)-(a,b)\| < \varepsilon$, if $|f(x,y)-L| < B(x,y) = |B(x,y)| < \varepsilon$,

and so by definition $\lim_{(x,y)\to(a,b)} f(x,y) = L$

□

Example;

Show that $\lim_{(x,y)\to(0,0)} \frac{x^2y}{x^2+y^2} = 0$

We have $f(x,y) = \frac{x^2y}{x^2+y^2}$ and $L = 0$

$|f(x,y)-L| = |\frac{x^2y}{x^2+y^2}| = \frac{x^2}{x+y^2}\ |y| \leq \frac{x^2+y^2}{x^2+y^2}\ |y| = |y|,\ \ \forall (x,y) \neq (0,0)$

$B(x,y) = |y|$

Notice $\lim_{(x,y)\to(0,0)} B(x,y) = \lim_{y\to 0}|y| = 0$.

By Squeeze Theorem, $\lim_{(x,y)\to(0,0)} f(x,y) = 0$

Note: add to the numerator and do not subtrat from the denominator because it will change the domain of the funciton.

💡 **Tip — Useful Inequalities for Upper Bound**

1. $|a+b| \leq |a| + |b|,\ \ a,b \in \mathbb{R}$
2. If $c \geq 0 \Longrightarrow a \leq a+c$

   Common use case, typically combined with Squeeze Theorem: $|x| = \sqrt{x^2} \leq \sqrt{x^2+y^2}$
3. $2|x\|y| \leq x^2+y^2$

Note: The Squeeze Theorem requires to have a limiting value L in mind.

Example:

Determine $\lim_{(x,y)\to(0,0)} \frac{x^2-|x|-|y|}{|x|+|y|}$

Try $y = mx$

$\lim_{x\to 0} \frac{x^2- |x| - |mx|}{|x|+|mx|} = \lim_{x\to 0} \frac{|x|-1-|m|}{1+|m|} = -1$

We now want to prove that $\lim_{(x,y)\to(0,0)} \frac{x^2- |x| - |y|}{|x|+|y|} = -1$

$| \frac{x^2-|x|-|y|}{|x|+|y|} + 1 = |\frac{x^2- |x|-|y| +|x|+|y|}{|x|+|y|}| = \frac{x^2}{|x|+|y|} \leq \frac{|x|(|x|+|y|)}{|x|+|y|} = |x|, \ \forall (x,y) \neq (0,0)$

$\lim_{(x,y)\to(0,0)} B(x,y) = \lim_{x\to 0} |x| = 0$

By Squeeze Theorem, $\lim_{(x,y)\to(0,0)} \frac{x^2-|x|-|y|}{|x|+|y|} = -1$